Woxsen University

School of Technology

B.Tech - Computer Science and Engineering (Artificial Intelligence and Machine Learning)

IDEATION VALIDATION REPORT

Idea Screening, Problem–Solution Fit Validation and Project Details

Title of the Project:

VisionQC: AI Visual Quality Inspection as a Service for Small Manufacturers in Telangana

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1. Opportunity and Problem

Opportunity area: quality control in small manufacturing units in Telangana (auto components, packaging and castings). It was chosen because the buyer is clear (the plant or quality head), the cost is visible (rejected lots, rework, lost orders) and the solution matches the team's AI and ML skills.

Problem in one sentence (without naming the technology): small plant heads lose orders to defects that escape manual end-of-line inspection, and they cannot afford an industrial inspection system.

Small plants depend on one or two inspectors who check large numbers of parts over long shifts. As attention drops, more defects reach the buyer, and the plant keeps no record of which defects occur, on which part or in which shift. Industrial machine-vision systems exist, but they are built for large plants and need a capital budget.

2. How the Ideas Were Generated

The ideation funnel from the course was followed: observe, frame, diverge, converge and validate.

Step	What the team did
Observe	Looked at where small plants lose money: quality, downtime and inventory. Quality has the clearest owner (the plant head) and the clearest cost.
Frame	Stated the problem in one sentence without naming the technology (Section 1).
Diverge	Generated candidate ideas in the same opportunity area and reshaped the core idea with SCAMPER (below).
Converge	Wrote five ideas in one sentence each and scored them on a weighted evaluation matrix (Sections 3 and 4).
Validate	Turned the uncertainties into hypotheses and planned tests with real plant heads (Section 7).

SCAMPER applied to the core idea

Prompt	Application to quality inspection
Substitute	Replace the tired human eye at the end of the line with a camera and a vision model.
Combine	Combine inspection with WhatsApp reporting and defect analytics in one service.
Adapt	Adapt computer vision used in large factories for small plants, using phones and low-cost cameras.
Modify	Minimise the setup: a phone on a stand instead of a fixed gantry with custom lighting.
Put to other use	Use the same approach across auto parts, packaging and castings by retraining per part.
Eliminate	Remove the capital purchase and installation team by charging per batch.
Reverse	Check at the source of the defect, not only at the end of the line.

3. Idea List

Five candidate ideas in the same opportunity area, each written as: for whom, what problem, what solution.

ID	Idea	For whom, what problem, what solution
A	VisionQC (phone-based inspection) Pitched in the Business Model Canvas	Plant heads lose orders to defect escapes; photo-based AI inspection sold per batch.
B	Edge camera box	Fixed-line plants need continuous inspection; hardware unit plus monthly software subscription.
C	Defect analytics dashboard	Plants with manual logs cannot see defect patterns; analytics on existing records, no cameras.
D	Remote inspectors	Plants without QC staff; human experts review photos remotely and send reports.
E	Predictive maintenance	Plants lose time to machine breakdowns; vibration sensors and alerts.

4. Evaluation (Screening) Matrix

4.1 Criteria and weights

The recommended criteria and weights from the course were used without change. Each criterion is scored from 1 (weak) to 5 (strong). The weights add up to 100%.

Criterion	Weight	Question
Desirability	25%	Do target users care enough about the problem and outcome?
Feasibility	20%	Can the team realistically deliver the solution with available capabilities?
Viability	20%	Is there a credible path to revenue, savings or sustainable value?
Uniqueness	10%	Is the idea meaningfully different from current alternatives?
Impact	15%	How significant is the potential benefit for users and the organisation?
Scalability	10%	Can the solution grow without cost and complexity rising at the same rate?
Total	100%	

4.2 Weighted scoring matrix

Criterion	Weight	A VisionQC (pitched in BMC)	B Camera box	C Dashboard	D Remote	E Predictive
Desirability	25%	4	4	3	3	4
Feasibility	20%	4	2	5	4	2
Viability	20%	4	3	3	3	3
Uniqueness	10%	3	4	2	2	3
Impact	15%	4	4	3	3	4
Scalability	10%	4	3	4	3	4
Weighted total	100%	3.90	3.30	3.40	3.10	3.30

Weighted score = sum of (rating × weight). For Idea A: 4×0.25 + 4×0.20 + 4×0.20 + 3×0.10 + 4×0.15 + 4×0.10 = **3.90**. The scores are team-agreed ratings. They are a decision aid, not proof, and are to be revised after validation.

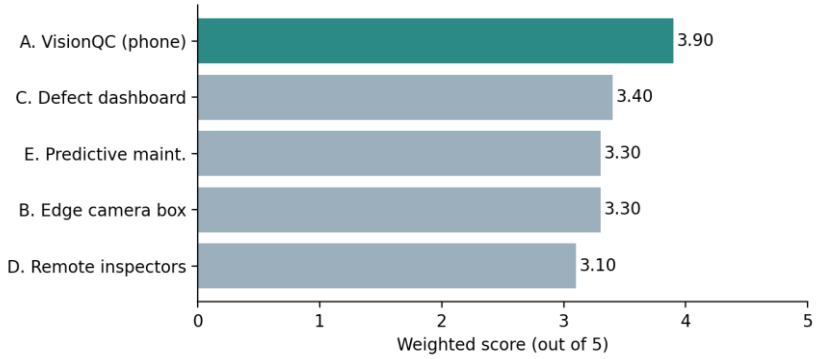


Figure 1: Weighted scores of the five candidate ideas

5. Selection of the Idea

Question	Answer
1. What won?	Idea A, VisionQC (phone-based inspection), with a weighted total of 3.90. This is the idea pitched in the Business Model Canvas.
2. Why?	The strongest criteria are Desirability (4), Feasibility (4) and Viability (4): the buyer has a visible need, the team can build a first version with phones and standard vision models, and it can be sold per batch from an operating budget.
3. What concerns us?	Uniqueness is only 3 because large vision vendors exist. Accuracy on varied parts and lighting is unproven. Idea C scored 5 on Feasibility and is the fallback if the camera setup proves difficult.
4. What must be tested first?	(1) Do plant heads treat defect escapes as a top-three problem? (2) Will they send real part photos? (3) Will they pay per batch from an operating budget?

Value proposition: for plant heads of small manufacturing units who lose orders to defect escapes, VisionQC catches defects at the line and reports them the same day using computer vision on a phone or low-cost camera, charged per batch so that no capital approval is needed.



Figure 2: VisionQC service flow

6. Business Model Canvas of VisionQC

The canvas describes the selected idea. It is a hypothesis about how the business creates, delivers and captures value, and it is updated with the evidence from Section 8. Prices are proposals to be validated.

Block	VisionQC
1. Customer Segments	Small auto-part, packaging and casting units in Telangana that recently had a rejected lot. Buyer: plant or quality head. User: line supervisor. Payer: plant owner.
2. Value Proposition	Catch defects at the line; same-day defect report on WhatsApp; defect records by part and shift; pay per batch with no capital budget; the model learns each plant's own parts.
3. Channels	Direct plant visits; industry associations and events; referrals from pilot plants; WhatsApp for reporting.
4. Customer Relationships	Free sample audit, then paid pilot, then monthly plan; WhatsApp support; quarterly review with the plant head.
5. Revenue Streams	Per-batch fee INR 500 (up to 200 parts), monthly plan INR 15,000 per line, one-time setup INR 10,000 (proposed).
6. Key Resources	Labelled defect image dataset; vision models; the founders' AIML skills; cloud compute; phones, cameras and stands.
7. Key Activities	Collect and label defect images; train and tune models; onsite setup; weekly reporting; customer support.
8. Key Partners	Industry associations; camera and stand suppliers; cloud GPU provider; freelance annotators; manufacturing advisors.
9. Cost Structure	Cloud compute, cameras, labelling, travel, tools and founder time (about INR 30,000 fixed per month; about INR 3,500 variable per line per month).

7. Validation Plan

Hypotheses are written before collecting data. The assumptions that are both high impact and high uncertainty, willingness to share photos and willingness to pay, are tested first.

Type	Hypothesis	Method	Success threshold
Customer	Small plant heads own the quality problem and decide inspection spending.	Customer interview	4 of 5 describe a defect incident in the last 3 months
Problem	Defect escapes cost money at least monthly (rejection, rework or complaint).	Interview (past behaviour)	3 of 5 report one at least monthly
Alternative	Plants cope with manual checks, sampling or re-checking after complaints.	Interview, observation	Current workaround recorded for all 5
Value	Plant heads will share part photos and use a same-day defect report weekly.	Concierge test	3 of 5 send 20 or more photos
Payment	At least some plant heads will pay per batch from the operating budget.	Pilot offer	1 of 5 agrees to a paid pilot

Participants and interview approach

- **Participants:** at least 5 real participants, such as plant owners or quality heads of small manufacturing units, or related small businesses with an inspection step. Each is recorded with role, business, date and method.
- **Concierge test:** the plant sends photos of parts and the team returns a manual defect report, before any model is built.
- **Ask (past behaviour only):** Tell me about the last time a defective batch reached a customer. What did you do instead? How often does it happen and what does it cost you? How do you inspect parts today? What have you already tried?
- **Do not ask:** “Would you use our AI?”, “Do you think this is a good idea?”, or any leading question.

Experiment Card

Field	Content
1. Assumption	We believe small plant heads will pay per batch for photo-based defect detection.
2. Test	To test this, we will interview five plant owners or quality heads, run a concierge photo test with each, and then offer a small paid pilot.
3. Target participant	We need evidence from small manufacturing units (auto parts, packaging, castings) and their quality heads.
4. Success threshold	We will consider it supported if 4 of 5 describe a recent defect incident, 3 of 5 send at least 20 photos and 1 of 5 agrees to a paid pilot.
5. Evidence collected	See Section 8.
6. Decision	Proceed, refine or pivot, with the reason, written after the evidence is in.

8. Evidence and Learning Summary

8.1 Evidence collected

Filled after each interview with real participants. Raw notes, photos or screenshots are attached as supporting evidence.

#	Participant (role, business)	Date, method	Last defect incident	Cost / frequency	Current workaround	Photos sent?	Would pay?
1	[ ]	[ ]	[ ]	[ ]	[ ]	[ ]	[ ]
2	[ ]	[ ]	[ ]	[ ]	[ ]	[ ]	[ ]
3	[ ]	[ ]	[ ]	[ ]	[ ]	[ ]	[ ]
4	[ ]	[ ]	[ ]	[ ]	[ ]	[ ]	[ ]
5	[ ]	[ ]	[ ]	[ ]	[ ]	[ ]	[ ]

8.2 Learning summary

Evidence	Pattern	Contradiction	Decision
What did we hear or observe?	What repeated across participants?	What challenged our assumptions?	Proceed, refine or pivot, and why?
[ ]	[ ]	[ ]	[ ]

8.3 Threshold check

Success threshold	Required	Result
Participants describing a recent defect incident	4 of 5	[ ] of 5
Participants reporting a monthly rejection, rework or complaint	3 of 5	[ ] of 5
Participants sending 20 or more photos	3 of 5	[ ] of 5
Participants agreeing to a paid pilot	1 of 5	[ ] of 5

8.4 Conclusion

“We believed that small plant heads would pay per batch for photo-based defect detection. We tested it by [interviews, photo concierge test and pilot offer with five participants]. We observed [result]. Therefore, we will [proceed / refine / pivot].”

9. Project Overview and Objectives

Sections 1 to 8 are the ideation and validation work. Sections 9 to 14 give the basic project details that support it.

Proposed solution. VisionQC is an AI visual inspection service delivered in three steps. A phone or low-cost camera photographs each part near the line. A computer-vision model, trained on that plant's own parts and defect types, marks defects such as scratches, cracks, misfits or missing features. The plant head receives a defect log on WhatsApp the same day, with a simple dashboard of defect counts by part, shift and type.

Concierge first. Before any model is built, the first version is delivered manually: the plant sends photos, the team reviews them and returns a defect report. This tests whether the plant sees value in the output, which is the riskiest assumption, before the technology is automated.

Objectives of the project:

- Generate several candidate ideas in one opportunity area and screen them with explicit criteria and weights.
- Select one idea with a recorded rationale and write its critical assumptions as testable hypotheses.
- Validate with at least five real participants and decide whether to proceed, refine or pivot.
- Describe the business with the Business Model Canvas, including prototype maturity, risk, organisation, retention and workflow.
- Estimate the basic economics and plan the first four weeks of implementation.

10. Prototype Maturity, Team and Organisation

Item	Detail
Current stage	Idea, with a concierge photo test planned next.
Technology readiness	TRL 2 (concept formulated) today; target TRL 4 (prototype validated on one part type with real photos) in four weeks.
Next milestone	Five interviews, concierge photo test, then a basic detection model on one part type.
Team	Mohammed Danish Khairati: product and machine learning. Emmanuel: customer outreach and operations. Rehan, Sriman and Aahil: [roles to be confirmed by the team].
Skills gaps and advisors	Manufacturing domain expertise and a field technician for setup; faculty guide Prof. Geeta Tripathi and industry association contacts act as advisors.
Customer retention	Free sample audit, setup on one line, same-day first report and weekly summary; monthly plan with quarterly review. KPIs: monthly churn, defect-catch rate reported by the plant, report open rate and plant-head satisfaction.
Workflow and supply chain	Photos and defect examples → labelling and training → test on plant parts → daily or weekly reports. No physical inventory; cameras from local distributors, cloud compute and labelled images. Bottleneck: quality and quantity of labelled images, and the team's time.

11. Risk Management

Likelihood and impact are rated Low (1), Medium (2) or High (3); the score is their product.

ID	Risk	L	I	Score	Mitigation and trigger
R1	Accuracy drops on new part types or lighting	2	3	6	Retrain per plant; keep human review. Trigger: accuracy below the agreed level in the pilot.
R2	Slow buying decisions; pilots do not convert	3	2	6	Free sample audit, then a per-batch fee from the operating budget. Trigger: no paid pilot after four weeks.
R3	Privacy of production images	1	3	3	NDA, local storage option, access controls. Trigger: plant raises a concern.
R4	Large vision vendors compete	2	2	4	Compete on price, speed and local support. Trigger: a pilot is lost to a larger vendor.
R5	Small team bandwidth	2	3	6	Narrow focus to one part family; use advisors. Trigger: missed weekly milestones.
R6	Camera and hardware variability	2	1	2	Standard phone stand and checklist. Trigger: repeated poor image quality.

12. Revenue Model and Unit Economics

The figures are illustrative assumptions to be validated, not market data. They show how the model behaves if the pricing in the canvas is accepted.

Item (per line, per month)	Assumed value (INR)
Revenue: monthly per-line plan	15,000
Variable cost: cloud compute	1,500
Variable cost: camera and stand (spread over use)	1,000
Variable cost: support and reporting time	1,000
Contribution per line	11,500 (about 77%)
Fixed monthly cost (tools, travel, freelancers)	30,000
Break-even number of lines	30,000 / 11,500 = about 2.6, so 3 lines

If a customer stays 12 months, contribution per line over its life is about INR 1,38,000 under these assumptions. Customer acquisition cost is not yet known and will be measured in the pilots. At INR 500 per batch, about 30 batches a month on one line equals the monthly plan.

13. Implementation Roadmap

Week	Action	Output
1	Write the problem in one sentence; interview five plant owners about the last defect incident; name the buyer and budget line.	Interview notes and refined problem
2	Concierge test: plants send part photos; the team returns a manual defect report.	First proof of value
3	Train a basic detection model on the collected images for one part type.	Working prototype (target TRL 4)
4	Offer a paid pilot on one line; redraw the canvas using the evidence.	Pilot decision and updated canvas

Requirements: a laptop with GPU, phone cameras and stands, a labelling tool, cloud credits, and access to small plants through contacts and associations.

14. Pitch Summary and Conclusion

Part	Content
Hook	One rejected lot can wipe out a small plant's month, yet inspection still depends on a tired human at the end of the line.
Problem	Defects escape, orders are rejected, and industrial vision is built for large plants.
Solution	VisionQC: phone-based AI inspection with same-day WhatsApp reports, billed per batch.
Business	The plant owner pays per batch or per line per month; low capital and low cost to start.
Evidence	[results from Section 8]
Ask	Introductions to small manufacturing units and one paid pilot line.

Conclusion. Of five candidate ideas in small-plant quality control, VisionQC scored highest at 3.90, mainly on desirability, feasibility and viability. Its weakest point is uniqueness, and its main uncertainties are whether plants will share photos and pay per batch. The Business Model Canvas shows a low-capital, software-first service that fits the team's AIML skills, and the illustrative economics suggest it becomes viable at about three lines. The final decision to proceed, refine or pivot will be taken from the interview evidence in Section 8.

References

- Course material: From Ideas to Evidence, Idea Screening and Problem–Solution Fit Validation (student assignment deck).
- Course material: Creativity Tools and Ideation Techniques: From Idea Generation to Selection.
- Course material: Customer Discovery and Business Model Validation.
- Course material: Ideation and Business Model Design masterclass, Woxsen University, School of Technology.